Why do we use imaginary numbers in AC circuit analysis with capacitors and Inductors?

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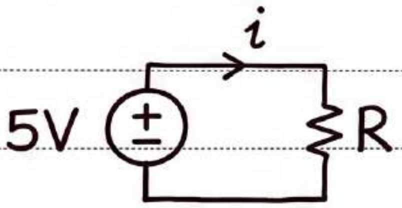
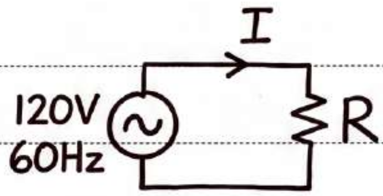
Why do we use imaginary numbers in AC Circuit analysis?

In DC, when having only a resistor: ratio between V and I stays the same at any time t $(R = V/I)$, hence V and I are in phase. Same thing occurs in AC when we have only a resistor.



$$i = \frac{V}{R}$$
$$\varnothing = 0$$



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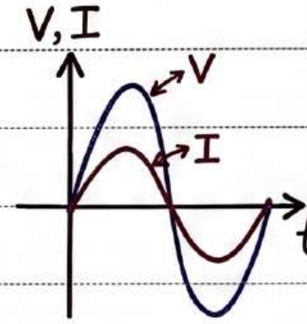


...With capacitors or inductors, this isn't the case...

$$i = C\frac{dv}{dt} \qquad v = L\frac{di}{dt}$$

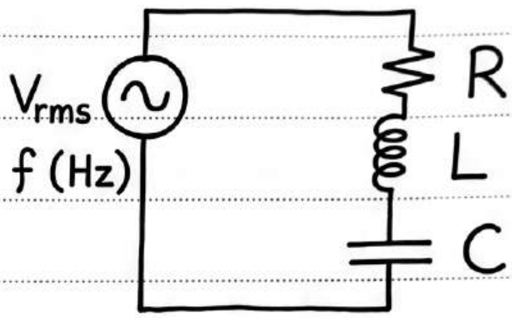
Clearly not in phase
when using sinusoidal
functions (sin, cos)

$v = \sin t$

$C = 1 \Rightarrow i = \cos t = \sin(t + 90°)$

Now, we can apply KCL here, but that would be messy.

Applying KVL:



$$V_S = V_R + V_L + V_C$$

$$V_S = V_{rms} \cos(t)$$

$$V_R = V_{Rmax} \cos(t)$$

$$V_L = V_{Lmax} \cos(t + 90°)$$

$$V_C = V_{Cmax} \cos(t - 90°)$$

$$V_{rms} \cos(t) = V_{Rmax} \cos(t) + V_{Cmax} \cos(t - 90°) + V_{Lmax} \cos(t + 90°)$$

→ For a simple circuit, this equation is too difficult to deal with. You can use sine and cosine rules

$$\sin(\alpha \pm \beta) = \sin\alpha \cos\beta \pm \sin\beta \cos\alpha$$

$$\cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha \sin\beta$$

Now to find any unknown (like V_{Rmax}), you would require using trig identities and experience math trauma.

OR, you can use Euler's formula:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

→ any trig expression can be rewritten as exponential.

Also! see: $1 + i = \sqrt{2} \cos(45°) + i\left(\sqrt{2} \sin(45°)\right) = \sqrt{2}\, e^{i(45°)}$

there is always a way to write complex number in this format.

~~For a capacitor. If we say $V = I/2\pi C$, but if we ignore phase for a moment and assume the equation :~~

Where the hell did this come from?

let's say Voltage across a component is $V_{comp} = A\cos(t + \phi_{comp})$, and we want to find the current across it. We can't just multiply V_{comp} by a constant to get the equation of I_{comp} sure multiplying by (-1) can phase shift by 180°, but we don't even know if I is shifted by 180°. So what do we do?

let's ~~see~~ observe this beauty for a second to see if it can benefit us :

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$2e^{i30^\circ} = 2\cos(30^\circ) + 2i\sin(30^\circ)$$

multiply both :

$$e^{i\theta} \times 2e^{i30^\circ} = 2e^{i\theta + i30^\circ} = 2e^{i(\theta+30^\circ)}$$

$$2e^{i(\theta+30^\circ)} = 2\cos(0+30^\circ) + 2\sin(0+30^\circ)$$

by multiplying two functions, we got a new function that has a magnitude of and phase shifted 30°. Surely we can use this in our main problem of $\boxed{V_{comp}}$ and $\boxed{I_{comp}}$ to find $\boxed{I_{comp}}$ across a capacitor or an inductor.

You can also use euler's formula to find the math identity of any sinusoid function.

For example:

$$\cos(2\theta) = ?$$

we can just place that as the angle in euler's formula $\Rightarrow$ $e^{2\theta i} = \cos(2\theta) + i\sin(2\theta)$

Hey! this is the same function we wanted to find its Value !

How can we find it? we can evaluate $e^{2\theta i}$ and find its real component, and that real component would equal $\cos(2\theta)$

$$e^{2\theta} = e^{2(i\theta)} = (e^{i\theta})^2 = (\cos\theta + i\sin\theta)^2$$

$$= \cos^2\theta + 2i\cos\sin\theta - \sin^2\theta \leftarrow i^2 = -1$$

$$= \underbrace{\cos^2\theta - \sin^2\theta}_{\text{real component}} + \underbrace{i\,2\cos\theta\sin\theta}_{\text{imaginary component}}$$

$$\boxed{\text{So, } \cos 2\theta = \cos^2\theta - \sin^2\theta}$$

Also, if we set imaginary components equal to one another: $\boxed{i\sin 2\theta = i\,2\sin\theta\cos\theta}$

Now let's apply that to the capacitor equation : $i = C\frac{du}{dt}$

note: to avoid confusion with current (i), imaginary symbol will be j. $j = \sqrt{-1}$

→ if $U = A\cos(\omega t + \phi)$, and $C = 1$ (for simplicity)

So, $i = -\omega CA\sin(\omega t + \phi)$

let's instead say:

$U \overset{(Re)}{=} A e^{j(\omega t + \phi)}$ → makes no sense, but the real component is our voltage

$= A\cos(\omega t + \phi) + jA\sin(\omega t + \phi)$

So, by the end we will see what the real component is

$$i = \frac{du}{dt} = \frac{d}{dt}\left(Ae^{j(\omega t + \phi)}\right) = j\omega C\,\underbrace{Ae^{j(\omega t + \phi)}}$$

So, we can say, $\boxed{i = j\omega C \cdot \overline{V}}$ *

$$i = j\omega C \times (A\cos(\omega t + \phi) + jA\sin(\omega t + \phi))$$
$$= \boxed{-\omega CA\sin(\omega t + \phi)} + j\omega CA\cos(\omega t + \phi))$$

real component is still i as we got normally by deriving $\left(\frac{du}{dt}\right)$ but ! It + gave us an equation of that is equivalent but doesn't involve differentiation.

$$\boxed{i = j\omega C\,\overline{V}} \Leftrightarrow \boxed{i = C\frac{du}{dt}}$$

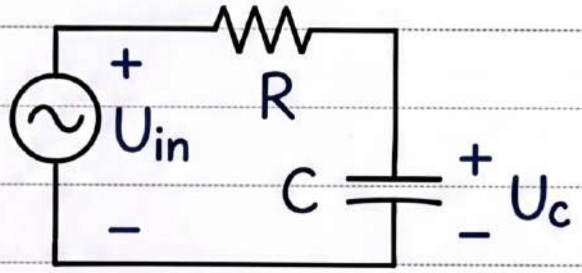
Now we can rearrange $i = j\omega CV$ to:

$$\frac{v}{i} = \frac{1}{j\omega C} = X_c$$

which is just the impedance of the capacitor to current.

Notice: $R = \frac{U}{i}$, and $X_c = \frac{U}{I}$ and the j of the imaginary numbers accounts for a phase shift of 90° just like cos and sin.

Now let's put a real example:



applying KVL:

$$U_{in} = U_c + U_R$$

$$U_R = iR = C \cdot \frac{dU_c}{dt} \cdot R = RC \cdot \frac{dU_c}{dt}$$

$$U_{in} = U_c + RC \cdot \frac{dU_c}{dt}$$

we know U_c would be of the form $U_c = A\cos(\omega t + \theta)$, but instead of substituting that, which wouldn't be fun, we can use the earlier function we came up with.

$$U_c = Ae^{j(\omega t + \theta)}, \text{ and } \frac{dU_c}{dt} = j\omega U_c$$

So, $U_{in} = U_c + RC(j\omega U_c) = U_c(1 + j\omega RC)$

and we finally get: $\boxed{U_c = \frac{1}{1 + j\omega RC} \cdot U_{in}}$

$\longrightarrow$ cont.

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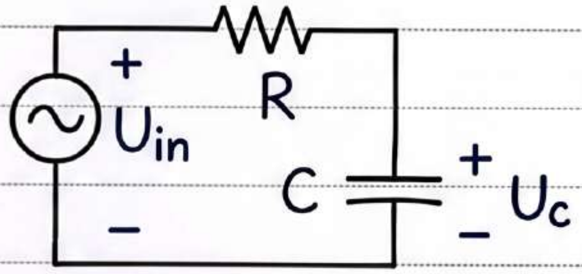

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cont.

For simplicity, $\omega = C = R = 1$

$$U_c = \frac{1}{1+j} \cdot U_i = \frac{1}{1+j} \cdot \frac{1-j}{1-j} \cdot U_i$$

$$= \frac{1-j}{1-j^2} \cdot U_i = \frac{1-j}{1+1} \cdot U_i = \left(\frac{1}{2} - j\frac{1}{2}\right) \cdot U_i$$

$$= \boxed{U_c = \frac{1}{2} U_i - j\frac{1}{2} U_i}$$

so, if $U_i = 3\cos(t + 10°)$

we can use euler's formula, $U_i = 3e^{j(t+10°)}$

note: while putting in mind that the actual voltage of $U_i = 3e^{j(t+10°)}$ is its real component.

$$U_c = \frac{\sqrt{2}}{2} e^{j(-45°)} U_i$$

$$U_c = \frac{\sqrt{2}}{2} e^{j(-45°)} \cdot (3e^{j(t+10°)})$$

$U_c = \frac{3\sqrt{2}}{2} e^{j(t-35°)}$, but of course we only care about the real component.

$$\boxed{U_c = \frac{3\sqrt{2}}{2} \cos(t - 35°)}$$

this result is how imaginary numbers help us.

Now:

After studying please solve problems on AC circuit analysis and use with it Euler's Formula to let consepts sink in.